MUSICAL ALGORITHMS

21M.370

Digital Instrument Design

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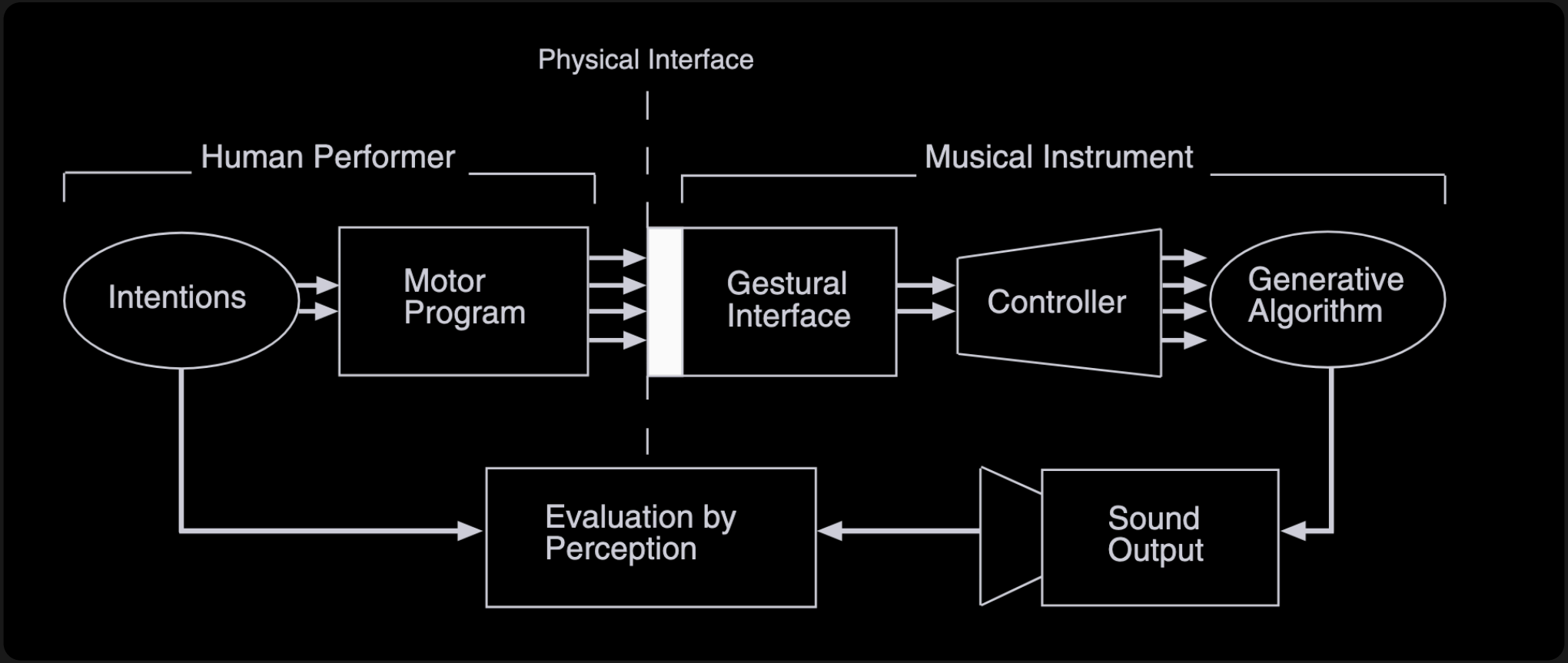
Problems and Prospects for Intimate Musical Control of Computers

David Wessel, Matthew Wright; Problems and Prospects for Intimate Musical Control of Computers. Computer Music Journal 2002; 26 (3): 11–22.

- immense timbral freedom of computers
- ‘one-gesture-to-one-acoustic-event’
- Low ‘‘Entry Fee’’ with No Ceiling on Virtuosity
- 10ms latency requirement
- Control Gestures as Events and Signals
- Musical Features all instrument should have
- Metaphors for musical control
- Open Sound Control



David Wessel, Matthew Wright; Problems and Prospects for Intimate Musical Control of Computers. Computer Music Journal 2002; 26 (3): 11–22.

different types of mappings

- one-to-one mapping: one gesture controls one sonic event
  - trigger a note
  - use a knob to control filter cutoff
- one-to-many
  - violin bow - controls volume, timbre, other
  - MPE controller: X:pitch, Z:volume, Y:timbre
- many-to-one
  - violin volume
  - filter cutoff: knob position, keytracking, lfo, envelope

Types of gestures

Discrete Events

- keyboard press
- Any note on event
- percussive strike

Semi-discrete

- continuous change, then static value
- Knob positions

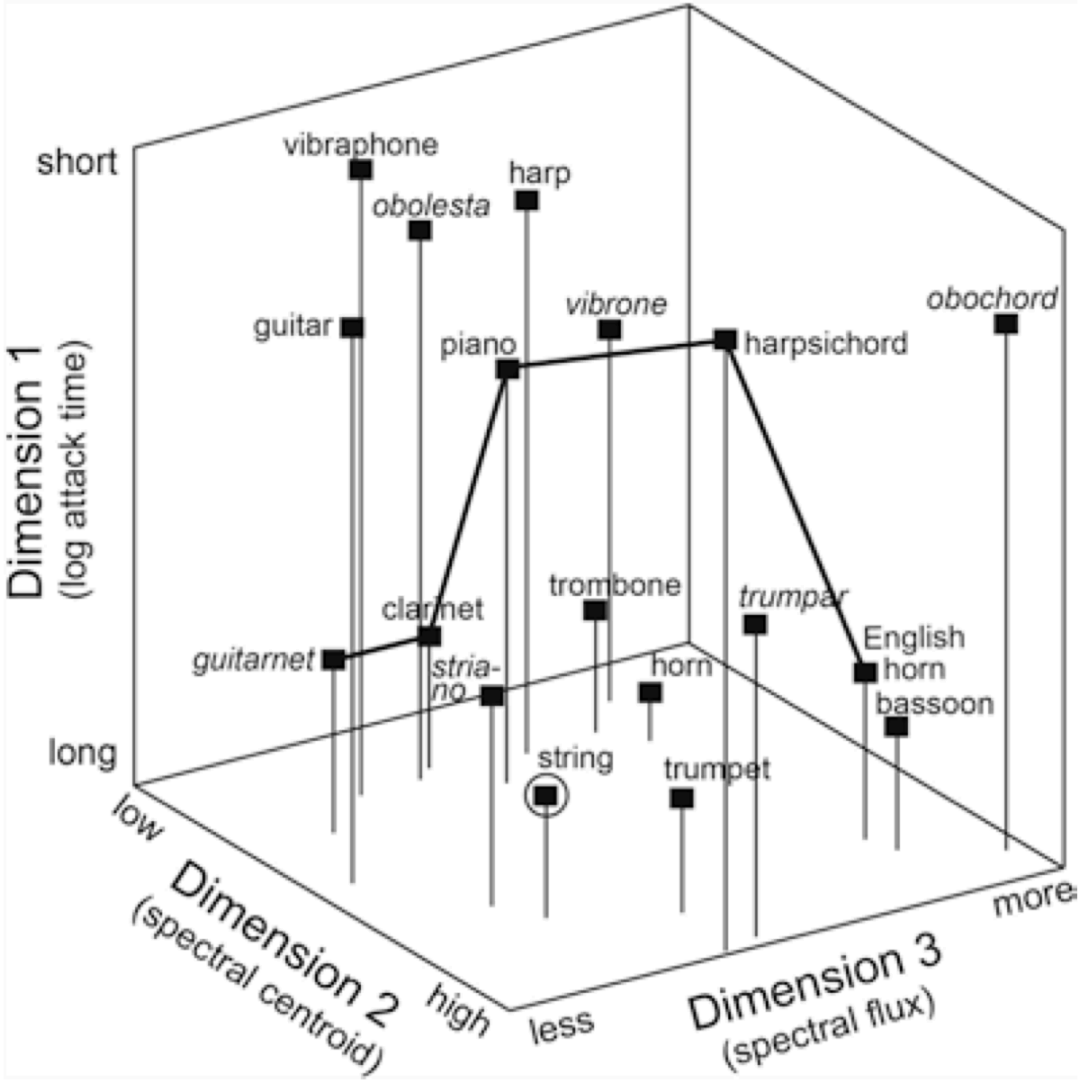
Continuous gestures

- gestural value only valid during the gesture itself
- change is the point of interest
- violin bow
- hand gesture in space
- gestural control of a potentiometer

Proposed gestural metaphors

- spatial metaphors
- drag and drop
- scrubbing
- dipping
- catch and throw

spatial metaphors



S. McAdams. K. Siedenburg et al. (eds.), Timbre: Acoustics, Perception, and Cognition,
Springer Handbook of Auditory Research 69, 2019.

dipping

- the computer constantly generates musical material via a musical process, but this material is silent by default
- performer controls the volume of each process
- each musical event can be precisely timed, regardless of the latency or jitter of the gestural interface
- its rhythm is not dependent on the performer in an event-by-event way

more general algorithms

- euclidean sequencing
- geometric functions to generate events or control streams
- terrain traversal
- markov models
- shaped / constrained randomness
- what other ideas?

Metaphors

DJs

- start with source material
- transforming material
- think compositionally over time
- disconnect between gesture and steady sound
- do things hidden
- preparation
- scratching
- timbral control
- catch and throw
- responsible for the whole mix

Piano

- start with a piano score
- gestural commitment
- gestual detail
- not a lot hidden
- fairly consistent timbre

Electric Gtr

- gtr, amp, effects pedals
- delay and time based effects as part of the sound

Production

- loop based performance
- mixing as performance
- production as key element
- short clips arranged in time
- visual metaphors
- humanizing as a conscious act

OPTIONS FOR NOBBY

- Find a way to implement rotation
- Decay time and amplitude for voices
- change the way that synth sequeces are generated